

Monodisperse SiO2 Microsphere OD600 Calibration Curve

Cross-Calibration of a Recirculating-Loop Inline Photometer Pair and a Benchtop Spectrophotometer — Standard Operating Procedure (SOP)

Document version: v1.0 | Material: Tomicro monodisperse SiO2 microsphere suspension (density 2.3 g/cm³, solids content 2.5%, mean particle diameter 1 μm, refractive index 1.46 @589 nm, 25°C) | Hardware: 100 mL beaker recirculating loop, peristaltic pump, two inline photometers, benchtop spectrophotometer (BioDrop or equivalent)

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1. Purpose

This document provides a standard operating procedure for building a "particle concentration vs. OD600" regression curve by cumulative titration of a known-concentration monodisperse silica (SiO2) microsphere suspension. The goal is to simultaneously calibrate two inline photometers installed in a recirculating loop, using OD600 values measured on a benchtop spectrophotometer (standard 1 cm pathlength) as an external reference anchor, so that the resulting conversion factors can be applied to real-time optical monitoring in a high-cell-density perfusion bioreactor (HFM) system.

2. Materials and Equipment

Item	Specification / Notes
Monodisperse SiO2 microsphere stock	Tomicro, density 2.3 g/cm ³ , solids content 2.5% (25 mg/mL), mean particle diameter 1 μm, refractive index 1.46 (589 nm, 25°C), total volume 10 mL
Deionized (DI) water	For preparing the working stock and system circulation fluid
100 mL beaker	Serves as the reservoir for the recirculating loop
Magnetic stir bar and stir plate	Keeps the suspension homogeneous; prevents microsphere sedimentation
Silicone tubing	Peristaltic-pump grade; connects beaker, pump, and the two photometers
Peristaltic pump	Recommended flow rate ~60 mL/min
Inline photometers	Two units, connected in series within the recirculating loop
Benchtop spectrophotometer	BioDrop or equivalent, capable of a 1 mL sample at 1 cm pathlength, 600 nm
Micropipettes and tips	1000 μL and 100 μL ranges

Data log	Paper log sheet or spreadsheet (see Section 8 for recommended columns)
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3. Principle and Underlying Calculations

3.1 Converting Mass Concentration to Particle Concentration

For a solid sphere of diameter $d = 1\ \mu\text{m}$ (radius $r = 0.5\ \mu\text{m}$) and density $\rho = 2.3\ \text{g/cm}^3$:

Single-particle volume $V = (4/3)\pi r^3 = 5.24 \times 10^{-13}\ \text{cm}^3$
 Single-particle mass $= V \times \rho \approx 1.20 \times 10^{-12}\ \text{g}$ ($\approx 1.2\ \text{pg}$)
 Stock particle concentration (25 mg/mL) $\approx 2.08 \times 10^{10}$ particles/mL
 Total particles in 10 mL stock $\approx 2.08 \times 10^{11}$ particles

3.2 Theoretical Scattering Efficiency (van de Hulst Anomalous Diffraction Approximation)

Using the manufacturer-supplied refractive index (1.46 @589 nm, 25°C), the van de Hulst anomalous diffraction approximation — valid for weak refractive contrast with particle size on the order of the wavelength — gives an estimate of the single-scattering extinction efficiency:

Relative refractive index $m = n(\text{particle}) / n(\text{water}) = 1.46 / 1.332 \approx 1.096$
 Size parameter $x = 2\pi \times n(\text{water}) \times r / \lambda \approx 6.97$ ($\lambda = 600\ \text{nm}$)
 $\rho = 2x(m-1) \approx 1.339$
 $Q_{\text{ext}} = 2 - (4/\rho)\sin(\rho) + (4/\rho^2)(1-\cos \rho) \approx 0.81$

 Single-particle extinction cross-section $\sigma = Q_{\text{ext}} \times \pi r^2 \approx 6.36 \times 10^{-9}\ \text{cm}^2$
 Theoretical estimate: 1 mg/mL SiO₂ concentration $\approx \text{OD}_{600}\ 2.3$
 (1 cm pathlength, single-scattering assumption; multiple-scattering saturation not accounted for)

Note: This is a single-scattering theoretical value. As OD approaches or exceeds 1.0, multiple-scattering effects will cause the measured value to fall below the linear extrapolation, an effect that becomes especially pronounced at high concentrations (see Section 9).

3.3 Cumulative Titration Concentration Model (Sample-then-Spike Method)

This protocol adds 1 mL of working stock, mixes thoroughly, and then withdraws a 1 mL sample, keeping the total system volume V constant ($V = 50\ \text{mL}$). Because withdrawing an aliquot from a homogeneous mixture does not change the concentration of what remains, the following recursion applies:

$C_n = (V \times C_{n-1} + C_{\text{stock}}) / (V + 1)$

 Closed form: $C_n = C_{\text{stock}} \times [1 - (V/(V+1))^n]$

 With $V = 50$, $C_{\text{stock}} = 2.5\ \text{mg/mL}$:
 Cycle 30: $C_{30} \approx 1.10\ \text{mg/mL}$ (theoretical $\text{OD}_{600} \approx 2.5$)
 Cycle 45: $C_{45} \approx 1.48\ \text{mg/mL}$ (theoretical $\text{OD}_{600} \approx 3.4$)
 Cycle 60: $C_{60} \approx 1.74\ \text{mg/mL}$ (theoretical $\text{OD}_{600} \approx 4.0$)

4. Safety Precautions

- Wear gloves and eye protection when handling the SiO₂ suspension to avoid splash contact with eyes and skin.
- If weighing out raw powder is ever required (outside the scope of this protocol, which uses only the liquid stock), work inside a fume hood to avoid inhaling dust.
- Before starting the peristaltic pump, confirm all tubing connections are securely fastened to prevent leaks or spray during operation.

- Follow the manufacturer's electrical safety guidelines when operating the spectrophotometer and inline photometers.

5. Reagent Preparation

5.1 Experimental Working Stock (2.5 mg/mL, 60 mL)

Combine **6 mL** of the original stock (25 mg/mL) with **54 mL** of DI water and mix thoroughly to obtain 60 mL of SiO₂ working stock at 2.5 mg/mL.

Material budget: This recipe uses 6 mL (150 mg) of the 10 mL original stock, leaving 4 mL (100 mg) in reserve for mid-run top-ups or a repeat experiment. The 60 mL working stock supports approximately 60 cycles of the "add 1 mL" step, matching the 45–60 cycle design recommended in Section 7.

Vortex or shake both the original stock and the working stock thoroughly before use, as the microspheres will slowly settle on standing.

6. Hardware Assembly and System Initialization

Step 6.1 — Assemble the recirculating loop

Connect, in order: 100 mL beaker → silicone tubing → peristaltic pump → inline photometer 1 → inline photometer 2 → back to the beaker. Place a magnetic stir bar in the beaker on a stir plate.

Step 6.2 — Establish the blank on both inline photometers

Add 50 mL of DI water to the beaker and start the peristaltic pump (~60 mL/min) to fill the entire loop and both photometers, confirming no residual air bubbles remain. Allow at least 2–3 full loop turnovers to circulate (at 50 mL total volume and 60 mL/min, one turnover takes roughly 50 seconds; wait about 2–3 minutes) until the signal stabilizes, then record the blank readings from inline photometer 1 and 2.

Step 6.3 — Establish the blank on the benchtop spectrophotometer

Withdraw 1 mL of DI water from the beaker and measure it on the benchtop spectrophotometer as its blank. This 1 mL aliquot may be discarded; to keep the volume bookkeeping consistent for later cycles, immediately replace it with 1 mL of fresh DI water so the beaker returns to 50 mL before beginning the titration in Section 7.

7. Experimental Procedure

Step 7.1 (Optional) — Preliminary mini-scan

If the actual scattering efficiency of the system is uncertain, first run a quick 4-point mini-scan: cumulatively add 2, 5, 10, and 20 mg of SiO₂ (corresponding to 80, 200, 400, and 800 µL of working stock), recording the inline photometer reading after each addition. This lets you quickly confirm whether the measured slope is on the same order of magnitude as the theoretical value from Section 3.2 (~2.3 OD·mL/mg) before committing to the full run. This step consumes only about 1 mL of working stock in total.

Step 7.2 — Standard operating sequence for each cycle (repeat every cycle)

1. With the peristaltic pump running continuously, pipette 1 mL of working stock (2.5 mg/mL) into the beaker.
2. Keep stirring and pumping; wait 2–3 full loop turnovers (roughly 2–3 minutes) to homogenize the concentration and let the signal stabilize.
3. Record the readings from inline photometer 1 and inline photometer 2 simultaneously.
4. Withdraw a 1 mL sample from the beaker, mix briefly, and immediately measure OD₆₀₀ on the benchtop spectrophotometer.
5. The cycle is complete; because 1 mL was added and 1 mL was withdrawn, the beaker volume returns to 50 mL and the next cycle can begin.

Step 7.3 — Repeat

Repeat Step 7.2 for a recommended total of 45–60 cycles. Per the model in Section 3.3, OD600 should approach 2.5 around cycle 30; the remaining cycles are used to observe the saturation behavior beyond the target concentration. If the 60 mL working stock is nearly exhausted while the signal is still rising without saturation, the remaining 4 mL of original stock (Section 5) can be diluted on the spot to continue the run.

8. Data Recording Template

At minimum, record the following columns for every cycle:

Cycle n	Cumulative stock added (mL)	Theoretical concentration C _n (mg/mL)	Inline photometer 1	Inline photometer 2	Spectrophotometer OD600	Notes
0	0	0	(blank)	(blank)	(blank)	System initialization
1	1	0.049				
...	...	...				
45–60	45–60	1.10–1.74				

Theoretical concentration formula: $C_n = 2.5 \times [1 - (50/51)^n]$ mg/mL

9. Quality Control and Troubleshooting

- **Microsphere sedimentation:** Keep the beaker under continuous stirring at all times. Measure withdrawn samples promptly, as microspheres will settle in a pipette tip or cuvette if left standing.
- **Consistency between the two inline photometers:** If the two units begin to diverge noticeably and persistently over the course of the run, this may indicate localized sedimentation in the tubing or a positional effect between the two sensors. Flag these points for later interpretation rather than discarding them outright.
- **Bubble management:** Starting the peristaltic pump can easily entrain microbubbles, which cause strong scattering artifacts and corrupt readings. Confirm the loop is fully degassed at startup.
- **Identifying the linear range:** As concentration increases, the measured OD will progressively fall below the linear extrapolation due to multiple scattering — this is an expected physical phenomenon. Use only the data points within the linear range for the formal regression, and flag the high-concentration, non-linear points separately as a reference for the instrument's effective upper measurement limit.
- **Batch-to-batch reproducibility:** If the microsphere batch or instrument hardware (light source, detector) is changed later, re-run this protocol to confirm the calibration curve is still valid.

10. Recommended Data Analysis

1. Plot the theoretical concentration C_n against each of the three datasets: inline photometer 1, inline photometer 2, and benchtop spectrophotometer OD600.
2. Visually identify the linear range for each dataset, and perform linear regression using only the in-range points to obtain each instrument's slope (conversion factor) and intercept.
3. Use the benchtop spectrophotometer (1 cm standard pathlength) regression as the external reference, and cross-calibrate each inline photometer's readings against it to establish an "inline reading → standard OD600" conversion equation.
4. Record the coefficient of determination (R²) and residual standard deviation for each regression as a quantitative measure of calibration quality.
5. Retain the raw readings for every cycle (not just the fitted regression coefficients) to support future comparisons when the strain, instrument, or microsphere batch changes.

11. References

1. Beal J, Farny NG, Haddock-Angelli T, Selvarajah V, Baldwin GS, Buckley-Taylor R, Gershater M, Kiga D, Marken J, Sanchania V, Sison A, Workman CT; iGEM Interlab Study Contributors. Robust estimation of bacterial cell count from optical density. *Communications Biology*. 2020;3:512. doi:10.1038/s42003-020-01127-5
2. iGEM Engineering Committee. *Measurement Kit — Particle Standard Curve with Microspheres Protocol*. International Genetically Engineered Machine (iGEM) Foundation. (Year-specific versions available on the Measurement/Protocols pages at igem.org)
3. van de Hulst HC. *Light Scattering by Small Particles*. New York: John Wiley & Sons; 1957. (Original source of the anomalous diffraction approximation for Qext used in Section 3.2)

The theoretical estimate in Section 3.2 is a single-scattering approximation based on the manufacturer-supplied refractive index. Actual scattering behavior may deviate from this estimate due to particle aggregation state, instrument geometry, and multiple-scattering effects; the measured data should be treated as authoritative over the theoretical estimate.